Vishal Khan

AI / Machine Learning Engineer

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LinkedIn | GitHub | AI Twin (live demo)

Chat with my AI Twin

Right to work in the UK. Part-time now, full-time from January 2027, then 18 months unsponsored. You need no sponsor licence to hire me until those 18 months end; after that I can switch to a Skilled Worker visa as a new entrant.

Four years in data roles, most recently at Teleperformance building Azure Databricks pipelines and a SQL Server migration, now building and testing LLM systems. I run a live RAG assistant on FastAPI, ChromaDB and Azure OpenAI with tool calling, per-source retrieval thresholds and an offline CI suite, and I audited the scorer of TRAIL, an agent-trace benchmark: a predictor that never opens a span scores **0.973** on GAIA against 0.183 for the paper's best model. MSc AI & Computer Science at Northeastern University London, finishing January 2027.

SELECTED ENGINEERING WORK

AI Digital Twin: Production RAG Assistant Live | Source

- OBJECTIVE: Answer recruiter and interviewer questions from grounded, cited evidence, including live GitHub data a static knowledge base cannot hold.
- APPROACH: FastAPI, ChromaDB and Azure OpenAI with **8** LLM-callable tools and three answer modes; every answer lists its sources underneath, and Interview mode also cites each claim inline. Source-aware retrieval applies per-source distance thresholds before preference boosting, so a preferred source wins near-ties but cannot pull an irrelevant chunk into the prompt; follow-ups fold the last two turns into the embedding query, holding subject continuity without a second LLM call. Auto-ingested GitHub READMEs are delimited as untrusted data against prompt injection. **React 19** and **TypeScript** front end streaming over SSE.
- RESULT: Live on Azure Container Apps since moving off Railway, at about **$4.70** a month, with GitHub Actions publishing the image to GHCR. **647 tests** run in CI on every push to main, with Azure OpenAI mocked so no API key is needed; CI also runs mypy strict on the app code, ruff and oxlint, and a test checks that every mode's prompt keeps the untrusted-data rule.

trail-scorer-audit: Auditing an Agent-Trace Benchmark Source

- OBJECTIVE: Test whether the two headline metrics of TRAIL, Patronus AI's benchmark of human-annotated agent traces (arXiv:2505.08638), measure how well a model finds and labels the errors in a trace.
- APPROACH: Before writing any code, fixed nine properties a competent scorer should have. The authors' scorer is fetched at a pinned commit, checked against a SHA-256 and run unmodified from those exact bytes, then scores seven predictors, from a perfect judge to programs that see only span identifiers. Every score and table in the README is rendered from a committed artifact, and CI fails if the two drift apart.
- RESULT: A predictor that never opens a span or reads the answer key scored **0.973** against 0.183 for the paper's best model (GAIA, joint accuracy) by emitting **129.1**× as many errors as the key holds, since neither metric divides by the number predicted. None of the nine held; seven failures move a published number.

EXPERIENCE

Teleperformance Global Business Pvt. Ltd. *June 2024 to Jan 2026*
Data Engineer

- Built and maintained **Azure Databricks** pipelines processing millions of telecom records (CDRs, billing, agent performance), serving a data engineering team of **50+** across multiple regions.
- Migrated on-prem SQL Server schemas to Databricks, improving query performance by ∼**20%** and cutting infrastructure cost by ∼**20%**.
- Owned over **20 pipelines** end to end on Databricks Workflows, translating ambiguous stakeholder requirements into scheduled jobs that retired **30+** recurring manual reports.

Avant Garde Clean Room & Engg. Solutions Pvt. Ltd. *Aug 2021 to Dec 2023*
Executive, Commercial & Sales Analytics

- Built ML sales forecasting models achieving **10-20% MAPE** on monthly revenue projections, used daily for commercial planning.
- Turned ad-hoc commercial questions into recurring Python and SQL reporting workflows the team relied on.
- Rebuilt revenue and collections reporting in Excel (macros, VBA), cutting the monthly cycle from ∼**2 hours** to ∼**30 minutes**.

Earlier: finance, accounts and taxation at Aryan Care Pvt. Ltd. (2021) and PJRJ Associates (2018-2020). Statutory filings, audit support, reconciliation.

TECHNICAL SKILLS

Languages	Python, SQL, TypeScript, JavaScript, R, HTML/CSS
LLM & Agents	RAG, prompt engineering, agentic systems, MCP, LangChain, tool calling, Azure OpenAI, prompt-injection defence, evaluation harnesses
Machine Learning	Deep learning, reinforcement learning, classification, regression, time-series forecasting, NLP, feature engineering, model evaluation
Libraries	PyTorch, TensorFlow, Keras, scikit-learn, SentenceTransformers, Pandas, NumPy, Statsmodels, Matplotlib, Plotly
Engineering	FastAPI, Pydantic, ChromaDB, REST, SSE streaming, React 19, Vite, Tailwind, Azure Container Apps, GitHub Actions CI, GHCR, Vercel, Railway, Streamlit
Testing & QA	pytest, vitest, Testing Library, mypy (strict), ruff, oxlint
Data & Cloud	Azure Databricks, Databricks Workflows, Azure, AWS, ETL pipelines, pipeline orchestration, SQL Server, MySQL, MongoDB, Neo4j
BI	Power BI, Tableau, Excel

EDUCATION

Northeastern University London, UK *Jan 2026 to Jan 2027*
MSc Artificial Intelligence & Computer Science | **95/100** in Programming for Data Applications

Liverpool John Moores University, UK *Aug 2023 to Aug 2024*
MSc Data Science

IIIT Bangalore, India [cert] *Mar 2022 to Apr 2023*
Executive PG Programme in Data Science

University of Delhi, India *Nov 2018 to Nov 2021*
BCom (Commerce)

Course: Ed Donner's *AI Engineer Core Track* (2026), covering QLoRA fine-tuning and a multi-agent RAG price estimator; I moved four of its agents onto Azure OpenAI.

RESEARCH

MSc Thesis: Reinforcement Learning for Dynamic Pricing in E-commerce *LJMU, 2024*

- OBJECTIVE: Test whether RL agents outperform rule-based pricing policies in a simulated e-commerce environment.
- APPROACH: Built a custom Gym-style environment on real e-commerce data, then trained and compared DQN, A2C and PPO agents across multi-episode simulations.
- RESULT: **DQN** produced the most stable profit curve; wrote up the comparative analysis as the dissertation.

Current dissertation (Northeastern London, due 2027): agentic AI, focused on agent reliability and evaluation tooling.